4 **(a)** **(i)** From the graph, $R = 80\ \Omega$, $T = 40\ \Omega$, [1]

$E = I R_T$

$3.0 = I(80 + 40)$

$I = 25 \times 10^{-3}$ A [1]

(ii) **(**When the temperature increases from 0 °C to 75 °C, the resistance of R increases slowly compared to resistance of T, which is decreasing more rapidly, thus,**)** <u>the effective (or total) resistance decreases at a decreasing rate</u>. [1]

{Students who do not describe how the <u>rate</u> of effective resistance changes will only get 0 m}

(iii) From the graph at 30 °C,

resistance of R = 55 Ω and resistance of T = 110 Ω

[1 m for both correct. Accept ± 5 Ω]

Using potential divider rule,

p.d. across T = 110/(110 + 55) x 3.0 V = 2.0 V [1]

(b) **(i)** Resistance of wire $= \rho L / A$

$= 1.20 \times 10^{-6} \times 1.20 / [\pi \times (0.200 \times 10^{-3})^2]$ [1]

$= 11.46\ \Omega$ (accept 3 or 4 sf) [1]

(ii) p.d. across XY, $V_{XY} = [11.46 / (11.46 + 1.0 + 5.0)] \times 9.0$

$= 5.907$ V [1]

At balance point, p.d. across XJ = p.d. across thermistor = 2.0 V **(from aiii)**

Therefore, $XJ / XY = V_{XJ} / V_{XY}$

$XJ / 1.20 = 2.0 / 5.907$

$XJ = 0.406$ m [1, ECF from part (aiii)]

(iii) When the temperature is at 75 °C, the resistances of R and T are equal, the <u>p.d. across thermistor will decrease (to 1.5 V)</u>. Hence, <u>the balance length XJ will be shorter.</u> [1]